

predicting continuous numerical value
which category

① Good fitting
span, not span

② Hospital diagnosis

classification → predicting categories as class

no. of hours	result
1	fail
2	fail
3	fail
4	pass
5	pass
6	pass

fail → 0
pass → 1
0.6
0.5, 0.4
-0.4

Linear R.
→ $b_0 + b_1x$
predict output
straight line

classify
0 and 1
predicts probability
probability curve

Logistic Regression

$y = b_0 + b_1x$
 $y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$

$z = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots + b_nx_n$

Suppose: age = 30, salary = 10 lakh, credit score = 750, loan amount = 10 lakh

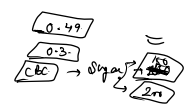
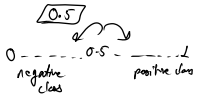
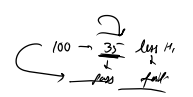
$z = 5, z = -3.5$

Sigmoid function

Input features → known z → Sigmoid function → probability $y \rightarrow A \in (0, 1)$
find class

$\sigma(z) = \frac{1}{1 + e^{-z}}$
0, 1

Decision Boundary



0 → negative
1 → positive

environment
→ index position of

fruits = ['apple', 'orange']

for i, fruit in enumerate(fruit):
print(i, fruit)

0 Apple
1 Orange
2 Mango

['Dog', 'Cat', 'Lion']

0 Dog
1 Cat
2 Lion

1 → Male
0 → Female

M → 0
F → 1
 $\begin{cases} M \rightarrow 1 \\ F \rightarrow 0 \end{cases}$

Age	Gender	Discrete (class)
25	M	0
35	F	1

